

Logic Explanation

Data Vortex :: AARUUSH'26 | Team SPARTANS

E1 - Platform Popularity

Filter out NULL platforms, GROUP BY platform, COUNT posts, sort descending.

SQL Techniques: GROUP BY

E2 - Most Engaged Posts (Top 10)

Calculate total_engagement = likes + shares + comments. Filter out posts with NULL likes. Order descending, LIMIT 10.

E3 - Average Engagement by Platform

GROUP BY platform, compute AVG for each engagement metric. Sum the averages for total. YouTube leads with 4044.54.

SQL Techniques: GROUP BY

E4 - Highly Shared but Poorly Liked

Simple WHERE clause filtering: shares > 1500 AND likes < 500. These posts have abnormally high sharing relative to their like count.

E5 - Users With Large Audiences (>40K followers)

Direct filter on follower_count > 40000, sorted descending.

M1 - Which Locations Generate the Most Engagement?

JOIN posts with users on user_id. COALESCE(likes, 0) handles NULLs. GROUP BY location, SUM engagement, rank descending. Los Angeles leads.

SQL Techniques: JOIN, COALESCE, GROUP BY

M2 - Do High Follower Users Get More Engagement?

Use CASE expression to bucket users into High/Low groups. JOIN with posts, calculate AVG engagement per post per group. Surprisingly, low-follower users get slightly more engagement.

SQL Techniques: CASE Expression, JOIN, COALESCE, GROUP BY

M3 - Most Active Users (Top 10)

JOIN posts with users. GROUP BY user_id, COUNT posts, SUM engagement. Order by post_count DESC to find the most prolific posters.

SQL Techniques: JOIN, COALESCE, GROUP BY

M4 - Platform Behaviour by High Follower Users ($\geq 30K$)

Filter users with $\geq 30K$ followers, exclude NULL platforms. GROUP BY platform, compute AVG engagement. Instagram performs best for high-follower users.

SQL Techniques: JOIN, COALESCE, GROUP BY

M5 - Detect Suspicious Engagement (Top 20)

Suspicious = shares > (likes + comments). This indicates artificial amplification or bot-driven sharing behaviour. 2,915 such posts exist.

SQL Techniques: COALESCE

H1 - Users With Abnormally High Engagement

CTE 1: Calculate per-user average engagement. CTE 2: Calculate overall average across all posts. Filter users whose avg > 2x overall. Result: 0 users exceed threshold (7044.63) - engagement is uniformly distributed in this dataset.

SQL Techniques: CTE, Window Function, JOIN, COALESCE, GROUP BY

H2 - Rank Users Within Their Location (Top 3)

Window function RANK() OVER (PARTITION BY location ORDER BY engagement DESC) assigns ranks within each location. Filter WHERE rank ≤ 3 to get top 3 per city.

SQL Techniques: CTE, Window Function, JOIN, COALESCE, GROUP BY

H3 - Exceptional Posts ($\geq 2x$ Platform Average)

CTE computes each platform's average engagement. Main query JOINS back and filters posts where engagement $\geq 2x$ their platform's average. These are statistically exceptional outliers.

SQL Techniques: CTE, JOIN, COALESCE, GROUP BY

H4 - Follower to Engagement Anomaly

Multi-level analysis: CTE 1 aggregates per-user totals. CTE 2 calculates the 90th percentile threshold using OFFSET. Filter: < 5K followers BUT engagement in top 10%. These are anomalously high-performing low-follower accounts.

SQL Techniques: CTE, JOIN, COALESCE, GROUP BY, Subquery/OFFSET

H5 - Identify Data Anomalies (Corrupted Dataset)

Uses the CORRUPTED dataset (not cleaned). CASE expression classifies each anomaly type. Checks 4 corruption patterns: negative likes (CAST to REAL), missing platform, missing text, HTML entities/tags via LIKE patterns.

SQL Techniques: CASE Expression

H6 - Most Suspicious High Impact Users

Three-condition filter using CTEs: (1) follower_count < 10K, (2) avg engagement > overall average, (3) MAX(CASE WHEN shares > likes) = 1 checks if ANY post has more shares than likes. These users are suspiciously high-performing despite low follower counts.

SQL Techniques: CTE, Window Function, CASE Expression, JOIN, COALESCE, GROUP BY